



Education: Why not a race to the top? [☆]

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ABSTRACT

This essay reminds us that more than five decades of research establish education as one of the best economic investments in terms of future economic growth and material well-being. Despite that, global education is facing both a crisis of access and equity and a crisis of quality and relevance. The essay emphasizes the need for a holistic transformation of education across four dimensions: learning to learn, learning to do, learning to live together, and learning to be. To effectively advance in that direction, it advocates for the need to transform schools, teachers, learning methods, and resources, including the harnessing of the digital revolution. The essay identifies a paradox of educational investment, revealing huge global disparities and calling for increased and better investments. It discusses the danger of poverty traps, where short-term gains impede long-term investments in education. Ultimately, it calls for a shift from a “race to the bottom” mentality to a “race to the top” approach, emphasizing the need for visionary leadership, a social movement and international cooperation to break free from poverty traps and advance towards sustainable development.

1. Introduction

It's been 50 years since the publication of [Psacharopoulos and Hinchliffe \(1973\)](#) seminal book *Returns to Education: An International Comparison*, where they rigorously demonstrated what most teachers and educators already knew both from intuition and from experience: while education is much more than just an economic investment, the fact is that, in simple economic terms, education pays.

Five decades of research – and the experience of many countries – have confirmed this time and again: investing in education is one of the best, if not the best investment a country can make to increase income and improve the material wellbeing of its population. And yet, paradoxically, we are far from investing enough in education.

2. A crisis of access and equity, a crisis of quality and relevance

We know that a person who cannot read and understand a simple text, a person without basic numeracy skills, without a basic understanding of the relation between cause and effect, or without basic social skills, is a person lacking the capacities for leading a successful life in today's world and for contributing to our collective well-being.

But despite what we might know about the importance of education,

the truth is that education, across the world, is in deep crisis. On the one hand, it is a crisis of equity and inclusion, as millions remain out of school; on the other, it is a crisis of quality and relevance, as many of those who do have access to schooling, are not even learning the basics.

The combination of longstanding poverty, increasing inequality, misguided policies, and the impact of the pandemic, are leaving a vast majority of children in less developed countries with so little education that only three out of ten would be able to read and understand a simple text by age ten ([World Bank, 2022b](#)). This means that seven out of ten persons in less developed countries – but also in many regions and neighborhoods of rich countries – are being left out of their right to learn and, therefore, left out of their productive participation in society and from the corresponding benefits.

Of course, education is much more than merely an economic investment with very high rates of financial return. As stated in the Secretary General's Vision Statement at the Transforming Education Summit ([Guterres, A, 2022](#)), education – their right to learn – must be understood holistically in terms of four key dimensions.

First, they must *learn to learn*. This starts with their ability to read and write, to identify, understand, and communicate clearly and effectively, and to develop numeracy, digital and scientific knowledge and skills. Education must also instill in them the curiosity, the creativity, and the

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capacity for critical thinking and to nurture social and emotional skills, empathy, and kindness. This is essential for developing their capacity to deal with complexity in an increasingly uncertain and interconnected world.

Second, they must *learn to do*. As the world of work undergoes rapid and fundamental changes, so must education change to prepare every person for the challenges of the future – including the green, the digital and the care economy – and offering them life-long learning opportunities both in formal and in informal education.

Third, and this is a bigger challenge for educational systems, they must *learn to live together*. In a world of increasing inequality, rising tensions, fraying trust, weakened democratic culture and a dramatic environmental crisis, education must help us to live better with each other and with nature. This has to do with ethics, equality, and justice; with civic responsibility, democracy, and human rights; with the respect, understanding and enjoyment of our rich human diversity; and, of course, with our capacity and active commitment as global citizens with the goals of a confronting poverty and inequality, of promoting respect for the environment and pursuing a fair and sustainable development.

Of particular importance here is developing unabashed respect for human rights and not just the tolerance, but the enjoyment of human diversity in all its forms. In particular, education must promote gender equality. This requires a gender-sensitive curriculum that promotes sex and affective education, addresses gender-based prejudice, norms, or stereotypes, empowers and equips learners to combat violence against women and sexually diverse persons, and ensure adequate sexual and reproductive health for all.

Finally, and this is something educational systems very often forget or underestimate, they must *learn to be*: The deepest purpose of education lies precisely in learning how to live well, instilling in learners the values and capacities to lead a meaningful life, to enjoy that life, and to live it fully. Education must expand every learner's potential for creativity and innovation; their capacity to enjoy and to express themselves through the arts; their awareness of history and the diversity of cultures; and their disposition for leading a healthy life, to practice physical activities, games, and sports.

Education, so understood, is a key enabler for every person's right to be an active, productive, and significant member of society. Education implies both their right and the responsibility to contribute to society, to be active citizens, to be productive and creative workers, to be caring family members and good friends, to be able to lead a meaningful life and to enjoy it. That is what education is supposed to produce: people capable of living together well.

3. Transforming schools, teachers and learning resources

In order to confront both the crisis of equity and inclusion and the crisis of quality and relevance, and to advance towards a holistic education and a truly learning society, education itself must be transformed.

4. More and better schools

Schools, for starters, must be available and accessible for all; but that is not enough. To be truly inclusive, schools must become safe, healthy, and stimulating learning environments that support equality and inclusion for all. We must confront and revert every kind of exclusion affecting those often marginalized or underserved, such as rural populations; low-income households; migrants, refugees, and displaced persons; and people experiencing discrimination on the grounds of disability, ethnicity, religion, sex, sexual orientation, or gender identity.

5. More and better teachers

Teachers, of course, are the cornerstone of education and they will remain the key to effective quality learning. But the teaching profession

shares in the crisis of education. We face not only a shortage of teachers – and of quality teachers – but also a weakened position of teachers in society and with respect to their students and their families.

To confront the shortage of quality teachers the answer should be apparent to any person trained in economics – or with basic common sense: we must improve their working conditions, ensuring fair pay and a fulfilling and engaging professional experience, so that teaching becomes and remains an increasingly attractive profession. But, again, that is not enough.

In order to transform education, teachers must transform themselves. They must go from passive and unilateral purveyors of information into active and stimulating guides in their students increasingly autonomous learning capacity. The authority of teachers will no longer depend on their having more information than their students – or on their punitive power – but, rather, on their capacity to instill the curiosity and the love for learning in their students and to lead them into the learning adventure of becoming themselves.

This has to do not just with the material conditions, but with the culture of teaching. We face two problems here. One has to do with the fact that being an active and innovative teacher who practices a learner centered pedagogy and must evaluate each student's individualized learning, is certainly a more difficult and more demanding challenge than that of merely repeating year after year the same packages of information, using the same textbooks – the same for all students – and then assessing their ability to repeat what they were taught. Teachers are usually not prepared, not supported, and certainly not incentivized for this difficult transition towards innovative and learner-centered teaching.

The second has to do with teachers' increasing sensation that they are losing authority and respect from their students and their families. This, again, has to do with the need to transform the teaching practice: in a world increasingly flooded with information, with easy access to knowledge and to stimuli of all kinds it should come as no surprise that today's students are neither impressed nor stimulated by traditional teaching methods. The old sources of authority do not work anymore: having more information or wielding the power of punitive assessments are no longer sufficient or adequate. We need to discover new sources of authority, respect, and teacher leadership. We need a new learner-centered pedagogy, a new teaching practice – and we need educational systems that promote and reward such new practices.

6. Harnessing the digital revolution for education

Like any other type of work, teaching depends not only on the personal capacities of teachers, but also on the tools and resources with which they perform their job. For too long, those tools have remained largely unchanged in the form of textbooks, notebooks, and blackboards. The digital revolution and the explosion of digital teaching and learning resources bring great promise to confront the inequities and inadequacies of current educational systems. The promise, however, also comes with great risks. For the digital revolution to really contribute to solve both the crisis of access and equity and the crisis of quality and relevance, it must be educationally and pedagogically harnessed.

As stated at the Transforming Education Summit, harnessing the digital revolution has to do with “the three Cs”: connectivity, capacity, and content. First, to close the equity gap, we must ensure connectivity and access to digital equipment: computers, tablets, smartphones, and others. In some places, access to electricity will be a precondition for digital connectivity.

Second, we must ensure there is sufficient capacity in teachers, principals, and education administrators to effectively leverage digital tools for promoting creative, evidence-based, and increasingly independent learning in their students.

And third, we must ensure the production and open distribution of high quality, curriculum-relevant digital content and resources so that they are truly available to all learners and teachers. These are typical

public goods which require a significant effort and a high fixed cost to be produced, but once produced, they can be widely used by teachers and students everywhere with very little or no additional cost. However, if left to the market, such resources would become artificially scarce and quite expensive, thus remaining inaccessible to teachers and poor students in many contexts. That is why we must transform digital teaching and learning resources into free and open global public goods, so that their financing, design, production, and distribution are organized so as to guarantee free and open access of teachers and learners everywhere.

7. Transforming the content and method of education

Finally, the content and method of education must also be up to the task and the challenges of today and tomorrow. Content must be relevant and open ended; it must be contextual – both local and global – and build on the needs, cultures, and traditions of a learning society. Foundational learning is essential, but the definition of foundational must evolve beyond literacy and numeracy in line with the changing world to also include digital skills, creative and critical scientific thinking, and socio-emotional skills.

Education needs to move from the traditional model of rote-learning – still prevalent in many places – to a more active approach, based on cooperation, collaboration, and solidarity, through activities such as project-based learning, collaborative methods, group problem-solving, critical thinking, and inquiry-based learning focused on contemporary issues. Assessment should also be formative to reflect these pedagogical goals in ways that promote meaningful growth and learning for all students.

8. Inequality and the paradox of educational investment

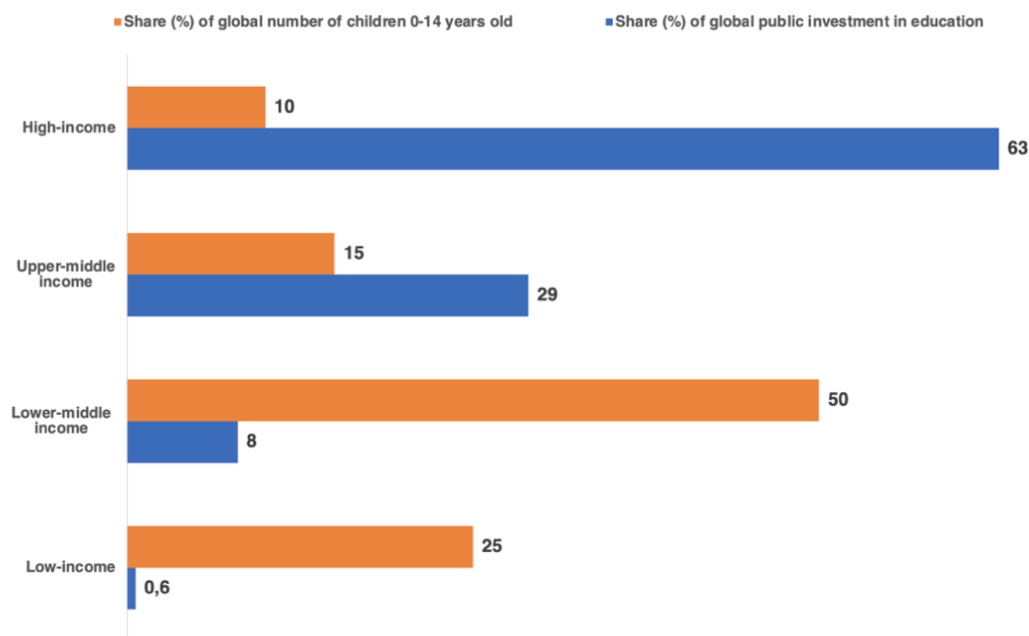
We know what it takes to transform education: more and better schools; more and better teachers; connectivity, capacity, and the content necessary to harness the digital revolution; and the promotion of a

much more relevant curricular content and a learner centered pedagogy. And we need it at scale. The right to education – if it is to be truly a right and not merely an aspiration – must be real for every person in the world – especially every young person. And yet, with almost 250 million children and youth out of school today (World Bank, 2022a) we are far from reaching that goal.

We might think that the challenge is too big to tackle. According to the Global Education Monitoring (GEM) Report, today, low- and lower-middle-income countries face a \$97 billion annual financing gap to achieve SDG4, the goal of universal quality education by 2030 (World Bank, 2023). It is true that \$97 billion is undoubtedly a large number, but we would be wrong to assume it is out of our reach. Today, we invest over US\$5 trillion per year in education in the world (World Bank, 2022a). That is about 6% of global GDP. The \$97 billion gap represents about 2% of that total, which means that the paradox of educational investment is, also, a paradox of unequal investment.

There are stark inequalities within countries, with many of those most in need of free and good quality public education are the ones receiving less effective access and support. But there is also a glaring global inequality of educational investment. Even though most children in the world live in low- and lower-middle-income countries, most of the world's educational investment is concentrated in high-income countries, which account for 63% of global investment in education but attend only 10% of the school-age population (using the 0–14-year-old population as a proxy). Next, we have upper-middle-income countries, with 29% of global educational investment and 15% of the school-age population. Very different is the situation of lower-middle-income countries, which, with only 8% of global investment, must serve 50% of the world's school-age population. Finally, low-income countries try to educate 25% of the world's school-age population with only 0.6% of global investment in education, which represents merely 0.03% of global GDP (World Bank, 2022a). In very simple terms, as of today, three quarters of the world's school-age population receives less than ten percent of global educational investment.

Share of global investment in education and of 0-14 year old global population by national level of income



UNESCO, World Bank: Education Finance Watch 2022, UN Population Division data

This unequal distribution of educational investment translates into a global reproduction of educational inequality, as the amount of resources each country can invest per school-age person greatly diverges. Roughly, by 2020, per capita spending in education was over \$8,000 per year in high-income countries, about \$1,000 in upper-middle-income countries, only \$300 in lower-middle-income countries and merely \$50 per year in low-income countries (World Bank, 2022a). On average, poor countries invest one dollar per week per school-age person. Such educational inequality, in turn, reproduces social, economic and political inequality.

In the low and lower-middle income countries, the challenge of educational investment can only be solved if the national effort is substantially complemented by international cooperation. For countries that are investing from \$50 to \$300 a year per person, even doubling their national effort – as they should – would not be enough. The international community must step in, reverse current trends, and make the difference.

In most countries, however, this investment can and should be financed with national resources, because it makes sense to do so. Investing in education should be seen not only as a moral and political imperative – which it is – but it should also be understood as a sensible economic investment. As I said at the beginning, there is plenty of evidence that “education pays”. The World Economic Forum recently summarized such evidence arguing that a single dollar invested in education at the primary and secondary level yields an estimated \$2.50 in additional gross lifetime earnings in lower-middle income countries, and as much as \$5.00 in lower-income countries – that is a 500% return on investment. And these are only the private returns. If we were to add the economic impact of indirect benefits of education, we would find, for example, that every \$1 spent on early childhood development interventions yields \$13 in economic returns (World Economic Forum, 2022).

But it is not only a question of investing more in education – which we certainly must – but also a question of investing better, that is, more equitably and more efficiently. As a simple example of how investing better can make a large difference, Angrist et al. (2023) make the case for what they consider two highly cost-effective educational interventions: one is structured pedagogy, which they define as a coherent package of inputs like textbooks, lesson plans, teacher training, and ongoing coaching of teachers that work together to improve in-class teaching; the other is teaching according to learning levels rather than age or grade. They show that, if they were to be deployed uniformly to reach 90% of the students in lower and lower-middle income countries, these interventions would cost on average \$18 per student per year or \$7.6 billion annually, generating \$65 in benefits for every \$1 spent. Their results show that increasing education expenditure by just 6% could increase learning by 120%, if directed towards these highly cost-effective interventions (Angrist et al., 2023).

So, yes, the figures that would need to be invested for every potential learner in the world to have access to a decent education are large, but they are not that large as to grant the assumption that we cannot do something about it. Yes, we could, that is if we really wanted to.

9. The danger of poverty traps: a race to the bottom

Why is it, then, that countries do not invest more in education? Why do not they invest a higher proportion of their GDP and their national budget? Why do not they increase their investment per student and per school age person? Why is there such strong opposition for a progressive revamping of tax systems that could increase the tax to GDP ratio and open more fiscal space for the financing of education?

To answer these questions, we need to understand that there is a very strong relation between the style of development a country has and the kind of education that comes with it, and it is a relation which goes both ways.

Putting it in very simple terms, when a country is highly unequal and has a large supply of very cheap labor, it can find itself in what economists call a low-level equilibrium or poverty trap. Such a trap results from a typical vicious cycle. Market economies work through incentives, and so, the type of investments most easily attracted by the abundance of cheap labor are typically unsophisticated investments, with low capital intensity, low productivity, and little need for human capital. But still, these investments can be very profitable, not because they contribute with increasing productivity, but because of their continued access to low-cost human and natural resources. Insofar as labor remains cheap and abundant, there is no need to upgrade productive processes, to introduce more modern technology, or to increase productivity and then, there is no need for a better educated working force. With no need for an increasingly qualified labor force, there is little incentive in such an economy for raising taxes to finance education, which is perceived as a mere expenditure. Insofar as wages remain low, there would be little economic incentive to dedicate resources to increasing the human capital of workers.

The situation can be even more complicated when countries get entangled into the typical race to the bottom, where they deregulate the labor market and the exploitation of natural resources, devalue their currencies and grant generous tax exemptions to further reduce costs in order to attract foreign investment.

As (Acemoglu and Robinson, 2012) have argued – and many others before them – in countries where these extractive or low-productivity economies prevail, the institutional framework tends to be weak and the balance of economic and political power is significantly skewed towards the upper echelons of income and wealth, which again tend to oppose the kind of progressive tax increases that would be necessary to finance universal quality education.

Education is the only way out of a poverty trap, but poverty traps curtail the capacity to invest in education, even though in the long run and for most of the population, such an investment would make social and economic sense. Short term profits, however, will not allow it.

The combination of the interests of global capital with those of the local elites conspire against significant progressive tax reforms capable of widening the fiscal space and, especially, against those reforms aimed at curtailing the capacity for eluding and evading taxes at a global level. As stated by the UN Secretary General, such transformations will require effective international tax cooperation in order to address tax evasion, aggressive tax avoidance, money laundering and illicit financial flows, and to improve and build confidence in tax systems (Guterres, 2023).

10. Why not a race to the top?

In order to achieve sustainable development, we must solve the educational paradox: we must transform education, but in turn, in order to transform education, we must also break free from perverse poverty traps and race to the bottom strategies. We need to reverse gears, we must provoke a race to the top.

This will take vision and, more than that, it will take a movement capable of altering the balance of power for a country to break free from these poverty traps, embarking on a true race to the top capable of taking advantage of the virtuous cycle of fair and sustainable development: increasing wages, increasing productivity, expanding and improving education, making a sustainable use of natural resources and strengthening political institutions. In the race to the top, education broadly understood emerges both as a key instrument, and as an objective in itself.

CRedit authorship contribution statement

Garnier Leonardo: Conceptualization, Investigation, Writing – original draft, Writing – review & editing.

Declaration of Competing Interest

The author declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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